AST1410 MESA Practical 1

Gurman Sachdeva
1007896314

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The tutorial followed in this practical used MESA to evolve a 15 solar mass star through to the zero-age-main-sequence phase (to the point in time when hydrogen burning ignites in the core). The condition used to stop the simulation here was $L_{\mathrm{nuc}}/L > 0.99$, meaning that 99% of the star's luminosity is generated through nuclear fusion. This was specified in the submitted `inlist_project` file. This file was updated to ignore the luminosity ratio condition and instead stop the simulation when the mass fraction of H1 in the star's centre dropped below a threshold. Finally, this inlist file was copied to another, `inlist_load`, used to load the results of the previous simulation and continue running beyond the main sequence, with wind properties in the RGB and AGB phases specified, and the simulation was manually stopped at around 600 steps. The resulting $\log L/L_{\odot}$ vs. $\log T_{\mathrm{eff}}$ data is plotted below in Figure 1. The plot was created using `plot.py`, which is a modified version of the file of the same name included by default in `/star/work`. I think the plot shows the data over all simulations that were run (out to step 600) but I'm not entirely sure.

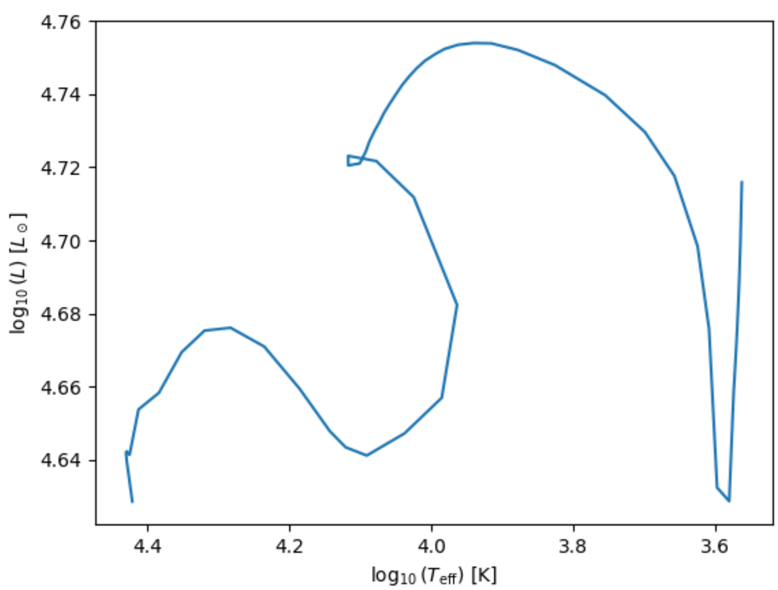

Figure 1: HR diagram showing evolution of 15 solar mass star in MESA from slighlty pre-MS to slightly post-MS.